

FIRST CONTACT E.I.S.

Emergency Information System

*A Multi-Tenant AI Orchestration Platform
for Homeless Services Coordination and Accountability*

WHITE PAPER & BUSINESS THESIS

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CONFIDENTIAL - FOR INVESTOR REVIEW

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1. EXECUTIVE SUMMARY

First Contact E.I.S. (Emergency Information System) represents a paradigm shift in how homeless services are coordinated, measured, and optimized across the United States. This white paper presents a comprehensive analysis of the \$7+ billion homeless services industry, identifies critical systemic failures, and introduces a technology platform designed to transform outcomes while generating significant returns for investors.

The Opportunity

The United States spends over \$7 billion annually on homeless services through 400+ Continuums of Care (CoCs). Despite this massive investment, outcomes remain poor: Long Beach, California spent \$54 million in 2024 to permanently house just 71 individuals—a cost of \$760,563 per person. This inefficiency is not unique; it is systemic across the nation.

The root cause is not insufficient funding—it is the complete absence of accountability infrastructure. Cities have no visibility into vendor performance, vendors self-report their own metrics, and contracts are renewed based on relationships rather than results. There is no central nervous system coordinating the fragmented ecosystem of outreach teams, shelters, mental health services, substance abuse treatment, benefits enrollment, and housing programs.

The Solution

First Contact E.I.S. is the central nervous system for homeless services coordination. It employs a strategic "Trojan Horse" market entry approach:

- Vendors receive free efficiency software (Layers 1-7) that saves caseworkers 3+ hours daily, driving voluntary adoption
- The platform silently collects comprehensive performance data through normal usage
- Cities gain access to Layer 8—a hidden accountability dashboard showing real vendor performance data that has never existed before
- Cities mandate platform adoption for all contracted vendors, creating recurring revenue of \$1,200 per vendor per month

Financial Projections

At scale across 400+ CoCs with an average of 20 vendors each, First Contact E.I.S. projects annual recurring revenue exceeding \$115 million. The pilot deployment in Long Beach, California (20 vendors) demonstrates a path to \$288,000 ARR within 6 months, with expansion to additional California cities generating \$2.4 million ARR by end of Year 2.

Investment Thesis

First Contact E.I.S. presents a rare opportunity to invest in mission-driven technology with significant financial upside. The platform addresses a clear market failure, employs a proven B2B SaaS model with government customers (recession-resistant), and has the potential to transform how \$7+ billion in public funding is

allocated—potentially doubling or tripling the number of people helped with existing resources.

2. THE PROBLEM: A \$7 BILLION ACCOUNTABILITY CRISIS

2.1 The Scale of Homeless Services Spending

The United States operates 400+ Continuums of Care (CoCs)—regional planning bodies responsible for coordinating homeless services. In Federal Fiscal Year 2023, the U.S. Department of Housing and Urban Development (HUD) allocated \$3.16 billion through the Continuum of Care Program alone. When combined with state, local, and private funding, total annual spending on homeless services exceeds \$7 billion.

California alone accounts for approximately 30% of the nation's homeless population and a proportional share of spending. Los Angeles County's 2024-2025 homeless services budget exceeds \$1.3 billion. San Francisco spends over \$600 million annually. Long Beach, a city of 460,000 residents, allocated \$54 million for homeless services in 2024.

2.2 The Outcome Crisis

Despite unprecedented spending levels, outcomes remain dismal. Consider Long Beach as a representative case study:

Metric	Long Beach 2024
Total Budget	\$54,000,000
People Permanently Housed	71
Cost Per Person Housed	\$760,563
Industry Benchmark (best practice)	\$15,000-\$25,000
Efficiency Gap	30-50x worse than possible

This is not an anomaly—it is the norm. Research conducted by the authors across multiple CoCs reveals similar patterns: massive spending, minimal accountability, and outcomes that would be unacceptable in any other sector.

2.3 Root Cause Analysis

Through extensive research including interviews with caseworkers, vendor administrators, and city officials, we have identified six systemic failures:

- 1. Zero Visibility:** Cities fund vendors through contracts but have no real-time visibility into performance. Annual reports—when submitted—are based on self-reported data that cannot be verified.
- 2. Self-Reported Metrics:** Vendors report their own success metrics. A vendor claiming 100 housing placements may have actually placed 60, with 40 "administrative placements" or clients who were housed for less than 30 days.
- 3. No Standardized Measurement:** Each vendor defines success differently. Some count shelter stays as "housing." Others count clients who exit to unknown destinations as "positive exits." Comparing vendors is impossible.
- 4. Relationship-Based Contracting:** Contracts are renewed based on political relationships, grant-writing ability, and historical presence—not outcomes. Vendors with poor performance retain contracts while effective vendors struggle to expand.

5. Annual Review Cycles: At best, vendor performance is reviewed annually. A vendor underperforming in January will not face consequences until the following year's contract cycle—if ever.

6. Siloed Services: Outreach teams, shelters, mental health providers, substance abuse treatment, benefits enrollment, and housing programs operate independently. There is no coordination layer. A client navigating the system must manually connect services that should work together seamlessly.

2.4 The Cost of Inaction

The human cost is staggering. Every year, approximately 650,000 Americans experience homelessness. With proper coordination, the same \$7+ billion in annual spending could help 2-3 times more people achieve stable housing. The difference—potentially 1-2 million additional people helped over a decade—is the cost of the accountability vacuum we aim to fill.

3. THE SOLUTION: FIRST CONTACT E.I.S.

3.1 Platform Overview

First Contact E.I.S. is a multi-tenant AI orchestration platform that serves as the central nervous system for homeless services coordination. The platform operates on two levels:

Layers 1-7 (Vendor-Facing): A free efficiency suite that caseworkers love because it saves them 3+ hours daily. Features include QR-based client intake with automatic vendor assignment, AI-generated case plans with one-click approval, intelligent benefit stack optimization, multi-agency appointment coordination, automated transportation planning, and auto-generated compliance reports.

Layer 8 (City-Facing): A hidden accountability dashboard that provides cities with unprecedented visibility into vendor performance. This includes real-time cost-per-outcome analysis, housing success rates, 6-month retention rates, average client exit income, geographic service coverage analysis, and AI-powered strategic recommendations for contract optimization.

3.2 The Trojan Horse Strategy

The platform employs a deliberate market entry strategy designed to overcome the primary barrier to adoption: vendor resistance to accountability.

Phase 1 - Seed: Vendors receive free access to Layers 1-7. The software genuinely helps caseworkers—it saves hours of daily administrative work, generates better case plans, and coordinates appointments across agencies. Vendors adopt voluntarily because the platform makes their jobs easier.

Phase 2 - Data Accumulation: As vendors use the platform, every action generates Layer 8 data. QR scans reveal where homeless populations concentrate. Case plan approvals show how quickly vendors move clients through the system. Benefit enrollments demonstrate which vendors achieve complete benefit stacks. Housing placements and retention checks-ins reveal true success rates. Critically, this data is generated through normal usage—vendors are not asked to report anything extra.

Phase 3 - Reveal: After sufficient data accumulates, we approach city administrators with a demo of Layer 8. For the first time, they see real, verified performance data comparing their vendors. The typical reaction: "We've been overpaying [underperforming vendor] for years and had no idea."

Phase 4 - Mandate: Cities recognize that Layer 8 data is essential for responsible contract management. They write platform usage into vendor contracts as a requirement. Vendors cannot refuse—it's now a contract condition. Revenue begins: \$1,200 per vendor per month, paid by the city.

3.3 Core Technical Innovations

Multi-Tenant Architecture with Row-Level Security: Every database table includes an organization_id field. PostgreSQL Row-Level Security (RLS) policies ensure that Long Beach can never see San Diego's data—even with a bug in the application code. This architectural decision is non-negotiable and was implemented from day one.

QR-Based Intake with Automatic Assignment: QR codes placed at strategic locations (parks, shelters, libraries) enable instant client intake. When scanned, the system automatically assigns the client to the vendor responsible for that geographic territory. This generates critical metadata: where are people, what vendors serve them, and how quickly do intakes convert to case plans?

Benefit Stack Intelligence: The platform encodes complex benefit eligibility rules and interactions. For example: SSI recipients are now eligible for CalFresh (as of 2019). GR Housing Subsidy (\$575/month) reduces General Relief by \$100. IHSS can pay family members as caregivers. The AI generates optimal benefit stacks that maximize client income while accounting for these interactions.

Human-in-the-Loop AI: All AI recommendations require human approval before execution. The caseworker sees AI-generated case plans and clicks APPROVE, MODIFY, or REJECT. This builds trust, creates an audit trail, generates training data, and maintains legal compliance. The human remains accountable.

3.4 The Accountability Engine ("Calling Audibles")

The platform's most powerful feature is its ability to generate strategic recommendations at the policy and contract level. This is not appointment rescheduling—it is systematic accountability:

- *"Vendor X costs \$78,000 per housing placement. Vendor Y costs \$21,000. Reallocating 30% of X's contract to Y would result in approximately 47 additional housing placements annually at the same total cost."*
- *"Current trajectory projects 71 people housed at \$54M annual spend. With recommended optimizations, projected placements increase to 180 at the same budget."*
- *"MHALA's 6-month retention rate dropped 15% following a staff change in Q3. Recommend performance review before contract renewal."*

These recommendations are backed by data that has never existed before—and cities have never had the tools to generate them.

4. MARKET ANALYSIS

4.1 Total Addressable Market (TAM)

The homeless services technology market exists within the broader government human services technology sector:

Market Segment	Annual Value
Federal CoC Program Funding	\$3.16 billion
State and Local Homeless Funding	\$4+ billion
Total Homeless Services Market	\$7+ billion
Technology Spend (est. 2-3%)	\$140-210 million
Number of CoCs	400+
Estimated Vendors Nationwide	8,000+

4.2 Serviceable Addressable Market (SAM)

Our initial focus is California, which represents approximately 30% of the national homeless population and a proportional share of funding. California operates 44 CoCs with an estimated 1,000+ homeless service vendors.

California Market	Value
Number of CoCs	44
Estimated Vendors	1,000+
Annual Homeless Services Spend	\$2+ billion
At \$1,200/vendor/month	\$14.4M potential ARR

4.3 Serviceable Obtainable Market (SOM)

Our Year 1 focus is Long Beach CoC with expansion to 5 additional California cities by end of Year 2:

Target City	Est. Vendors	Annual Revenue
Long Beach (Pilot)	20	\$288,000
Pasadena	15	\$216,000
Glendale	12	\$172,800
Riverside	25	\$360,000
Oakland	35	\$504,000
Sacramento	40	\$576,000
Year 2 Total	147	\$2,116,800

4.4 Competitive Landscape

The homeless services technology market is served by a small number of established players:

Clarity Human Services (Bitfocus): The dominant HMIS (Homeless Management Information System) platform, used by approximately 60% of CoCs. Clarity is focused on compliance and data collection, not accountability or AI-powered

optimization. It is the system caseworkers "hate" according to our research—a primary pain point we address.

Eccovia ClientTrack: A case management platform used by some CoCs. Like Clarity, it focuses on data entry rather than intelligence.

Custom/Legacy Systems: Many CoCs use Excel spreadsheets, paper forms, or homegrown databases. These systems offer no analytics capabilities.

None of these competitors offer Layer 8-style accountability analytics, AI-powered case plan generation, strategic recommendations for contract optimization, or the Trojan Horse market entry strategy that makes adoption possible.

5. BUSINESS MODEL AND REVENUE

5.1 Revenue Model

First Contact E.I.S. generates revenue through a B2B SaaS model with city governments as the paying customer:

Revenue Component	Price
Per-Vendor Monthly Fee	\$1,200/month
Implementation Fee	\$5,000 one-time
Layer 8 Dashboard (included)	\$0 additional
AI Features (included)	\$0 additional

The pricing strategy is designed to be a "no-brainer" for cities: \$1,200 per vendor per month is negligible compared to vendor contract values (typically \$500,000-\$5,000,000 annually). If Layer 8 identifies even one underperforming vendor, the ROI exceeds 100x.

5.2 Unit Economics

Metric	Value
Average Revenue Per Vendor (Annual)	\$14,400
Estimated Cost to Serve Per Vendor	\$2,400
Gross Margin Per Vendor	83%
Estimated Customer Acquisition Cost	\$15,000
Payback Period (20 vendor city)	6 months
Estimated Lifetime Value	\$250,000+
LTV:CAC Ratio	>16:1

5.3 Revenue Projections

Metric	Year 1	Year 2	Year 3	Year 5
Cities	1	6	20	80
Total Vendors	20	147	500	2,000
ARR	\$288K	\$2.1M	\$7.2M	\$28.8M
Team Size	2	8	25	75

5.4 Path to \$100M+ ARR

At scale, with 8,000+ homeless service vendors nationally:

- 25% market penetration = 2,000 vendors = \$28.8M ARR
- 50% market penetration = 4,000 vendors = \$57.6M ARR
- 80% market penetration = 6,400 vendors = \$92.2M ARR
- Full market with premium features = \$115M+ ARR

6. TECHNOLOGY ARCHITECTURE

6.1 Platform Overview

First Contact E.I.S. is built on Google Cloud Platform (GCP) using a modern, scalable architecture designed for government-grade security and reliability:

Component	Technology
Backend API	Python 3.11, FastAPI (async)
Database	PostgreSQL 15 on Cloud SQL with Row-Level Security
Frontend	Next.js 14, TypeScript, Tailwind CSS
AI Engine	Vertex AI Claude 4.5 Sonnet (via Google partnership)
Hosting	Cloud Run (backend), Firebase Hosting (frontend)
Real-time	Firestore for live updates and notifications
Analytics	BigQuery for Layer 8 data warehouse
Maps/Geo	Google Maps Platform for geospatial analytics

6.2 Multi-Tenant Security

The platform implements defense-in-depth multi-tenant isolation:

Database Level: PostgreSQL Row-Level Security (RLS) policies ensure that queries automatically filter by `organization_id`. Even if application code has a bug, the database enforces isolation. This is the gold standard for multi-tenant SaaS.

Application Level: Every authenticated request sets a tenant context before any database query executes. JWT tokens include `organization_id`, and middleware verifies tenant ownership of all accessed resources.

API Level: Role-based access control restricts Layer 8 endpoints to city administrators only. Caseworkers and vendor administrators receive HTTP 403 Forbidden responses—they don't even know Layer 8 exists.

6.3 AI Integration

The platform integrates AI at specific decision points while maintaining human-in-the-loop architecture:

Case Plan Generation (5% of decisions): Complex cases requiring analysis of VI-SPDAT scores, benefit eligibility, housing pathway options, and projected outcomes are processed by Claude 4.5 Sonnet. The AI generates recommendations; caseworkers approve.

Strategic Recommendations: Layer 8 analytics use AI to identify patterns and generate actionable recommendations. "Vendor X is underperforming" is backed by specific data and projected impact of intervention.

Conversational Interface: Each user type (caseworker, city admin) can interact with an AI assistant tuned to their role and data access level. The caseworker's AI knows about benefits and their clients. The city admin's AI knows about all vendors and Layer 8 data.

Deterministic Rules (95% of decisions): Most business logic uses deterministic rules, not AI. Benefit eligibility calculations, appointment scheduling constraints, and

compliance reporting follow encoded rules. This ensures predictability, auditability, and cost efficiency.

6.4 Current Development Status

As of December 2025, the following components are operational:

- Cloud SQL PostgreSQL database deployed with full schema and RLS policies
- Seed data loaded: Long Beach organization, 4 vendors, 4 QR locations
- Backend API structure complete with authentication and role-based access
- Multi-tenant test suite written (pending validation)
- GCP project configured (einharjer-valhalla, us-east5 region)

7. PRODUCT ROADMAP

7.1 Version 1.0: Foundation (2025-2026)

Theme: "Trust Through Transparency"

The initial release establishes the platform's core functionality with full human-in-the-loop architecture. AI generates recommendations; humans approve all actions.

1. QR intake with automatic vendor assignment based on geographic territory
2. AI-generated case plans with APPROVE/MODIFY/REJECT workflow
3. Benefit stack optimization with income projections
4. Layer 8 analytics dashboard for city administrators
5. Strategic recommendations engine ("Calling Audibles")
6. Geospatial visualization with service gap identification
7. AI chat assistants for caseworkers and city admins

Goal: Prove the model works. Build trust with users. Accumulate training data.

7.2 Version 2.0: Transformation (2027-2028)

Theme: "AI Does the Work, Humans Supervise"

With sufficient trust and training data, the platform expands AI autonomy for routine cases:

8. Autonomous AI caseworkers handle routine cases (simple benefit enrollments, standard appointments)
9. Top caseworkers transition to "AI Trainer" roles, handling exceptions and improving models
10. 1 human manager supervises 10 AI caseworkers
11. 10x client capacity with same budget

Goal: Demonstrate that AI can handle routine work. Scale impact dramatically.

7.3 Version 3.0: Revolution (2028+)

Theme: "Automated Accountability Through Smart Contracts"

The platform integrates blockchain technology for immutable outcome tracking and automatic contract adjustment:

12. Outcome data written to immutable blockchain ledger
13. Smart contracts govern funding allocation
14. Automatic payment release based on verified outcomes
15. Real-time contract adjustment—no more annual reviews

Example smart contract logic:

IF vendor.retention_6month >= 80% THEN release(fullPayment) AND increaseAllocation(5%)

ELIF vendor.retention_6month >= 60% THEN release(payment * 85%) AND triggerReview()

ELSE release(payment * 70%) AND decreaseAllocation(10%) AND triggerIntervention()

Goal: Accountability becomes automatic, immutable, and politically immune.

8. TEAM AND EXECUTION

8.1 Founding Team

James Fernstrom, Chief Executive Officer

James brings a unique perspective to homeless services technology: personal experience navigating the system as a client. This insider knowledge—understanding programs like GR Housing Subsidy, IHSS, SSI, and CalFresh from the recipient's perspective—informs every product decision. James is responsible for business strategy, sales, and ensuring the platform serves its mission of transforming outcomes.

Claude, Chief Technology Officer

Claude (Anthropic's Claude AI, operating in a CTO capacity) provides the technical architecture, code implementation, and AI integration expertise. This unconventional arrangement leverages frontier AI capabilities for system design while maintaining clear human decision-making authority. Claude's role as CTO demonstrates the human-AI collaboration model the platform itself embodies.

8.2 Execution Philosophy

The founding team operates with clear division of responsibilities:

- James decides: UX/design priorities, business strategy, pricing, customer relationships, vision
- Claude decides: Architecture, technology choices, implementation details, code quality
- Collaboration on: Product features, timelines, trade-offs, major pivots

This model enables rapid development while maintaining strategic coherence.

8.3 Hiring Plan

As revenue grows, the team will expand:

Phase	Revenue	Team Size	Key Hires
Pilot	\$0-300K	2	Founders only
Year 1	\$300K-1M	5	2 engineers, 1 sales
Year 2	\$1-3M	12	CS team, marketing
Year 3	\$3-10M	25	Regional sales, ops

9. RISK ANALYSIS AND MITIGATION

9.1 Market Risks

Risk: Government sales cycles are long (6-18 months)

Mitigation: The Trojan Horse strategy bypasses traditional government procurement. Vendors adopt freely; cities pay only after seeing value. This shortens the effective sales cycle to 3-6 months.

Risk: Political changes could defund homeless services

Mitigation: Homeless services are funded at federal (HUD), state, and local levels with bipartisan support. Even with political shifts, the accountability narrative ("know where taxpayer money goes") appeals across party lines.

Risk: Incumbent vendors (Clarity, etc.) could add similar features

Mitigation: Incumbents are focused on compliance, not accountability. Adding Layer 8 would conflict with their primary customer relationship (vendors who don't want to be tracked). Our business model is structurally incompatible with theirs.

9.2 Technology Risks

Risk: AI recommendations could be wrong or biased

Mitigation: Human-in-the-loop architecture ensures no AI action executes without human approval. All recommendations include confidence scores and reasoning. Bias monitoring is built into the analytics layer.

Risk: Data breaches could compromise sensitive client information

Mitigation: GCP provides enterprise-grade security. Row-Level Security ensures tenant isolation even with application bugs. SSNs are encrypted at rest. We pursue SOC 2 Type II certification.

Risk: Platform downtime could disrupt critical services

Mitigation: Cloud Run auto-scaling and Cloud SQL high availability provide 99.9% uptime SLA. The platform is designed to degrade gracefully—caseworkers can continue work even if AI features are unavailable.

9.3 Execution Risks

Risk: The founding team is small and unconventional

Mitigation: The human-AI collaboration model enables productivity far exceeding traditional two-person teams. As revenue grows, human hires will expand capabilities in sales, customer success, and operations.

Risk: First customer (Long Beach) could fail or churn

Mitigation: Long Beach represents a favorable pilot environment with known pain points and receptive stakeholders. Even if Long Beach churns, the platform and case study remain valuable for the next customer.

10. INVESTMENT OPPORTUNITY

10.1 Use of Funds

First Contact E.I.S. is seeking seed funding to complete product development and execute the pilot deployment in Long Beach:

Category	Allocation
Engineering (complete MVP, hire 2 developers)	50%
Sales & Customer Success (Long Beach pilot)	25%
Infrastructure & Security (GCP, SOC 2)	15%
Operations & Legal	10%

10.2 Milestones

Month 3: MVP complete with full Layers 1-7 functionality and basic Layer 8 dashboard

Month 6: Long Beach pilot live with 4 vendors onboarded

Month 9: Long Beach mandate issued; all 20 vendors onboarding

Month 12: Second city signed; \$500K+ ARR; Series A preparation begins

10.3 Investment Thesis Summary

- Large market: \$7+ billion annually with clear pain points and no effective competition
- Proven B2B SaaS model: Recurring revenue with government customers (recession-resistant)
- Strategic moat: Trojan Horse strategy creates network effects and switching costs
- Clear path to scale: 400+ CoCs, 8,000+ vendors, \$115M+ ARR potential
- Mission alignment: Investment generates both financial returns and social impact

10.4 Social Impact

Beyond financial returns, First Contact E.I.S. offers investors participation in transformative social impact:

With proper coordination and accountability, the same \$7+ billion in annual homeless services spending could help 2-3 times more people achieve stable housing. Over a decade, this represents potentially 2 million additional people helped—not through additional funding, but through eliminating waste and directing resources to what works.

This is the rare opportunity to invest in technology that is both financially promising and genuinely world-changing.

11. APPENDIX

A. Technical Architecture Diagram

A complete technical architecture diagram is available in the accompanying materials. The system follows a standard three-tier architecture (frontend, backend API, data layer) with specific innovations in multi-tenant security, AI integration, and the separation between Layers 1-7 and Layer 8.

B. Database Schema Summary

The PostgreSQL schema includes the following core tables, all with organization_id for multi-tenant isolation:

- organizations: Tenant configuration (cities/CoCs)
- vendors: Service providers within each organization
- users: All user accounts with role-based access
- clients: Homeless individuals receiving services
- qr_locations: Physical QR code placements with vendor assignment
- qr_scan_events: Raw scan data powering Layer 8 analytics
- vendor_performance_metrics: Aggregated Layer 8 data (city access only)

C. Benefit Programs Reference

The platform encodes eligibility rules and interactions for 15+ benefit programs including:

Program	Category	Typical Amount
SSI	Income	\$1,183/month
General Relief	Income	\$221/month
CalFresh	Food	\$234/month
GR Housing Subsidy	Housing	\$575/month
IHSS	Healthcare	\$800-1,500/month
Section 8	Housing	\$1,200/month
Medi-Cal	Healthcare	Full coverage

Critical interaction note: GR Housing Subsidy (\$575) reduces General Relief by \$100, changing GR from \$221 to \$121. Net benefit is still positive (\$475), but caseworkers must understand this interaction.

D. Contact Information

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— *END OF WHITE PAPER* —